

Clustering Models Analysis

Dataset Selection: `"sklearn.datasets import load_wine"`

Models: K-Means, K-Medoids, Agglomerative, GMM, DBSCAN

Algorithm Selection Analysis

- I ran multiple algorithms to see which could best capture the underlying structure in the PCA-transformed Wine dataset. K-Medoids and Agglomerative were straightforward with 3 clusters, and GMM gave slightly better silhouette scores but didn't drastically change the clusters. DBSCAN was surprising as it had the highest silhouette, but over 100 points were labeled as noise, which isn't really practical for most applications. (I was dead convinced that the results I was seeing were incorrect and that my model wasn't performing correctly. I re-wrote this part like 3 different times to make sure)
- The PCA features helped make the clusters more spherical and balanced, which favors centroid-based methods like K-Means or K-Medoids. DBSCAN was affected by the natural spread and outliers in the dataset.
- Overall, no algorithm completely contradicted the others, but the differences highlighted that each method captures slightly different data: K-Medoids kept everything, DBSCAN prioritizes dense regions and removed a ton of noise, and GMM accounts for probability distributions.

Cluster Quality Assessment

- I'm fairly confident the clusters we found are meaningful. Silhouette scores aren't perfect, but the clusters show clear separation in PCA space and statistical tests confirm significant differences across the features. Boxplots and radar charts show distinct chemical profiles for each cluster.
- Stability looks good too, and running K-Medoids with different initializations produced the same clusters, suggesting the structure is robust and not just an artifact of random initialization. There also weren't many (178 data points isn't that many in my book) data points to test against so the clusters were more well-defined.

- The combination of cluster profiles, statistical tests, and effect sizes makes me trust that these clusters reflect real groupings in the wine dataset, not noise or overfitting. It was also fairly easy to determine this from looking at the visual comparisons, especially the cluster maps.

Feature Analysis

- PCA shows that the first few components drive the separation that are related to alcohol, flavonoids, color intensity, and phenolics. PC1 had the strongest effect, followed by PC2, which aligns with the known chemistry of the wine classes.
- Features beyond the first 3 to 4 PCA dimensions contribute less and might be redundant for clustering. Scaling and dimensionality reduction helped emphasize the most informative features.
- For future experiments, including more chemically relevant features or removing low-variance dimensions could make the clusters even more distinct.

Practical Applications

- From a research standpoint, these clusters allow us to study chemical differences more systematically. Winemakers could use this to adjust fermentation or grape selection based on cluster profiles.
- In general, the segmentation gives extremely clear actionable insights into wine profiles, which could be translated into marketing, pricing, or product differentiation strategies.

Limitations & Future Work

- The biggest limitation is that clusters are based on PCA transformed data, so they don't directly map back to all original features. Outliers still affect cluster boundaries and can cause a lot of "noise" like we saw in the DBSCAN.
- Future work could include ensemble clustering, semi-supervised methods if partial labels are available, or even deep clustering to capture nonlinear relationships. More samples or additional chemical features would likely improve robustness and interpretability.
- Monitoring clusters over time is important if new wine samples are added. This would be a great pipeline for a winery or some kind of wine research center (I'd like to work there!) to keep adding to test against different features or chemical

compositions. I could also see the columns and features being expanded to contain some of the more scientific data to get more granular with what kind of results the stakeholders are looking for.

Ethical Considerations

- Clustering here is largely chemical and objective. I could reasonably see this dataset somehow combined/compared against demographic data to generate some kind of customer segmentation data. It could go as far as shipping “bad” or “undesirable” combinations and predictions to different markets to maximize profits. This is not inherently unethical per se, but knowing how some corporations and organizations operate it would NOT surprise me in the slightest.
- Transparency can be important. Certain customer segments are almost guaranteed to care more about specific features and flavor profiles for their “brand” of wine.
- There’s little privacy concern for this dataset specifically, but there are known wine makers who will go to the furthest extent to protect every bit of information about their crop down to how they water it. Should the data be expanded it could be reasoned to protect it and the evaluations for the sake of protecting the “recipe” and the brand.